PROBABILITY & STATISTICS · RESEARCH CASE STUDY

CAN PROBABILITY PREDICT A FLOOD?

Mumbai, 26 July 2005: 944 mm of rain in 24 hours.

THE PROBLEM

One day, one city, four numbers

944 mm

rain in 24 hours at Santacruz

74 mm

at Colaba, the same day, same city

481 mm

of it fell in just 4 hours

25 mm/h

what the city's drains were designed to handle

Was this a freak accident, or a measurable risk we could have planned for?

Sources: Gupta (2009), Disaster & Development 3(2); Rana et al. (2013), HESS Discussions 10:4709

THE RESEARCH WE BUILD ON

Three papers, three questions, our syllabus tools

Paper	Question	Method	Finding
Husak et al. 2007, Int. J. Climatology	Which distribution fits rainfall?	Gamma vs Weibull, goodness-of-fit test	Gamma fits ~98% of locations
Rajeevan et al. 2008, Geophys. Res. Letters	Are very heavy rain days (≥150 mm) rising?	Trend over 1901–2004, correlation	+6% per decade, significant at 99%
Roxy et al. 2017, Nature Comms	Are widespread extremes rising?	Trend with t-test, correlation	Extremes +75%, widespread events 3×

Our job: repeat their methods on real Mumbai data, using only distributions, hypothesis tests and regression.

Husak, Michaelsen & Funk (2007) 27:935 · Rajeevan, Bhate & Jaswal (2008) 35:L18707 · Roxy et al. (2017) 8:708

STEP 1 · FITTING A DISTRIBUTION · CHI-SQUARE GOODNESS OF FIT

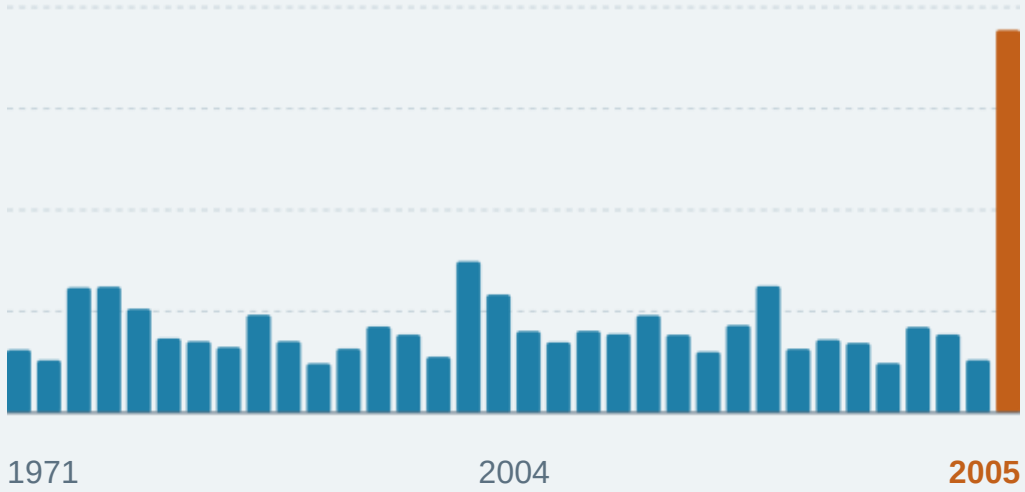
Which distribution does Mumbai’s rain follow?

Data: the **biggest one-day rain of each year**, 33 years. We fitted three distributions and ran a chi-square test (5 equal-probability bins, df = 5 − 1 − 2 = 2, critical value 5.99).

Distribution	χ^2	p-value	Verdict
Normal	7.45	0.02	Rejected
Weibull	6.24	0.04	Rejected
Gamma	1.39	0.50	Fits

Same answer as Husak et al.: **Gamma**. Rain is skewed right, so the symmetric Normal fails.

REAL DATA · SANTACRUZ, MUMBAI



Dashed lines every 250 mm. Before 2005 the record was **373 mm** (1965). Then came **944 mm**.

Data: NOAA GHCN-Daily, station IN012070800, 1971–2004 (33 complete years); 2005 value from IMD via Gupta (2009)

STEP 2 · BERNOULLI, GEOMETRIC AND BINOMIAL

The “100-year flood” is not once in 100 years

GEOMETRIC: WAIT FOR THE FIRST FLOOD

$$T = E[X] = 1 / p$$

Each year is a Bernoulli trial. A 100-year flood has p = 0.01.

BINOMIAL: AT LEAST ONE IN N YEARS

$$P(X \geq 1) = 1 - (1 - p)^n$$

One minus P(X = 0).

Years (n)	P(≥ one 100-yr flood)
10	9.6%
30	26.0%
50	39.5%
100	63.4%

A 30-year home loan has a 1-in-4 chance of seeing a 100-year flood.

Assumes years are independent with the same p each year

STEP 3 · GAMMA DISTRIBUTION · CONFIDENCE INTERVAL

What would the data have predicted in 2004?

1 · GAMMA FIT

Shape k = 12.3
Scale θ = 16.3
Mean kθ = 201 mm

2 · 95% CI FOR MEAN

$\bar{x} \pm t \cdot s/\sqrt{n}$
200.9 ± 2.04 × 10.7
179 to 223 mm

3 · T-YEAR RAIN

Find x where
P(X > x) = 1/T
from the Gamma CDF

Rain in one day	Return period
277 mm	10 years
358 mm	100 years
373 mm (1965 record)	~160 years
944 mm (2005)	~10 trillion years

The model passed the test, then called 2005 impossible.

Chi-square checks the bulk of the data. 33 years say little about the far tail.

Santacruz annual maxima 1971–2004 (n = 33), NOAA GHCN-Daily. Gamma parameters by maximum likelihood; t(0.025, 32) = 2.04

STEP 4 · BINOMIAL MEAN AND SPREAD

Did the model predict the right number of bad years?

$$E[X] = np \cdot SD = \sqrt{(np(1-p))}$$

X = years above a level, out of n = 33.

Level (Gamma)	Expected	Observed
2-yr · 195 mm	16.5	13
10-yr · 277 mm	3.3	5
50-yr · 335 mm	0.66	1

Both gaps are within about 1.2 SD: normal random variation. Yet the rarest levels are hit more often than predicted.

RESEARCH FINDING

< 1 year

Return period of Mumbai's 25 mm/h drain design.

Rana et al. (2013), 1951–2004 data. Exceeded more than once a year on average, so the drains are expected to overflow every monsoon.

Data: NOAA GHCN-Daily, Santacruz 1971–2004 (n = 33) · Rana, Bengtsson, Olsson & Jothiprakash (2013), HESS Discussions 10:4709

STEP 5 · REGRESSION, CORRELATION, T-TEST, F-TEST

Is extreme rain getting worse?

OUR TEST · ONE STATION, 33 YEARS

$\hat{y} = 1302.8 - 0.56 \times \text{year}$

r = −0.16. Slope t = −0.92 vs ±2.04, p = 0.37. 1971–70 vs 1984–2004: 207 vs 190 mm, t-test p = 0.43. F = 1.63 < 3.23, so equal variances is fine.

Fail to reject H_0: no significant trend.

THE BIG STUDIES · THOUSANDS OF STATIONS

+6% per decade · 3× more

Rajeevan 2008: very heavy rain days rising, 1901–2004 (99% significant). Roxy 2017: widespread extremes tripled since 1971.

“Not significant” ≠ “no trend”. One gauge and 33 years give low power, so a Type II error is likely.

Rajeevan, Bhate & Jaswal (2008), GRL 35:L18707 · Roxy et al. (2017), Nature Communications 8:708

THE ANSWER

NOT THE DATE. THE ODDS.

Distributions: chi-square picks Gamma for Mumbai's rain.

Geometric & binomial: a 100-year flood is 26% likely in 30 years.

Limits: the model fits the middle but missed 944 mm.

Trend: not significant at one station, but significant in large studies.

REFERENCES

Husak et al. (2007). Int. J. Climatology 27:935.

Rajeevan et al. (2008). Geophys. Res. Lett. 35:L18707.

Roxy et al. (2017). Nature Comms 8:708.

Rana et al. (2013). HESS Discussions 10:4709.

Gupta, K. (2009). Disaster & Development 3(2).

NOAA GHCN-Daily, station IN012070800.

Can Probability Predict a Flood? — Project Report

Probability and Statistics Lab Project · Vani Singh, Linga Chaitra · Nov 2, 2026

1. Summary

Probability cannot predict the date of a flood, but it can measure the odds. Picking the right model changes those odds by a factor of millions. We analysed 63 complete monsoons (1950–2024) from the Santacruz rain gauge in Mumbai. Every model was fitted only on 1950–2004 data and then asked about 26 July 2005. The Normal model put 944 mm of rain at about once in 5×10^{21} years, Gamma at once in 1.7 billion years, and Gumbel at once in 4.3 million years. The Generalised Extreme Value (GEV) distribution, which extreme-value theory recommends for annual maxima, put it at **about once in 300 years**: rare, but a risk that could have been planned for.

Key result	Value
Complete monsoons analysed	63 (1950–2024)
Best-fitting model for annual maxima	GEV, shape $\xi = 0.36$ (heavy tail)
Return period of 944 mm, fitted before 2005	308 years (95% CI 65 to 12,299)
Chance of at least one 100-year storm in 30 years	26%
Very heavy rain days (≥ 115.6 mm) per monsoon	Poisson, $\lambda = 3.62$
Mean monsoon total, 2010–24 vs 1990–2009	3,077 vs 2,269 mm (ANOVA $p < 0.001$)
Trend in the annual maximum	+7 mm per decade, not significant ($p = 0.29$)

All code, data and figures are in the project notebook `analysis/flood_analysis.ipynb`. The website turns the fitted model into an interactive flood-risk calculator.

2. Problem and design methodology

On 26 July 2005, Santacruz recorded 944 mm of rain in 24 hours. Mumbai's storm drains were designed for 25 mm per hour, and the previous Santacruz record was below 400 mm. The practical question is whether that day was unforeseeable, or whether a statistical model built on the data available in 2004 could have flagged it as a real risk.

We set four hypotheses before running any analysis:

1. **H-dist:** annual maximum rainfall follows a heavy-tailed extreme-value distribution, not a Normal one.
2. **H-count:** the number of very heavy rain days per monsoon follows a Poisson distribution.
3. **H-change:** rainfall has changed over time, tested on annual maxima, monsoon totals and heavy-day counts.
4. **H-predict:** a model trained on early years predicts later years within its stated uncertainty.

The workflow runs in six stages, each mapped to a theme in the brief:

1. **Clean.** Load the gauge record, apply a completeness rule, and fix documented recording errors (Section 3).
2. **Describe.** Compute central tendency, spread and shape, and plot histograms, box plots and time series (Section 4).
3. **Fit.** Fit Normal, Gamma, Gumbel and GEV models, check them with chi-square goodness of fit, and use the Binomial and Poisson distributions for predictions (Section 5).
4. **Test.** Run z, t, F, chi-square and ANOVA tests on the change hypotheses (Section 6).
5. **Model.** Fit simple and multiple linear regression for trends and drivers (Section 7).
6. **Validate.** Train on 1950–1990, test on 1991–2024, and explain every deviation (Section 8).

Every test uses $\alpha = 0.05$. Tail estimates carry 95% bootstrap intervals from 1,000 resamples.

3. Data and cleaning

Cleaning produced 63 complete monsoons from 76 calendar years, and it caught a recording error that halved the worst day on record. The source is NOAA GHCN-Daily station IN012070800 (Bombay/Santacruz): 19,808 daily rainfall values from October 1949 to August 2025, in mm. It has no duplicate dates and no negative values, but 7,914 calendar days are missing, mostly outside the monsoon after 2005.

Rule	Why	Effect
Keep a year only if ⩾ 104 of 122 June–September days (85%) are recorded	The annual maximum falls in the monsoon in almost every year. A season with big gaps may have missed its worst storm.	13 years dropped: 1971, 1972, 1973, 1975, 1976, 1977, 1983, 2000, 2009, 2014, 2015, 2022, 2025

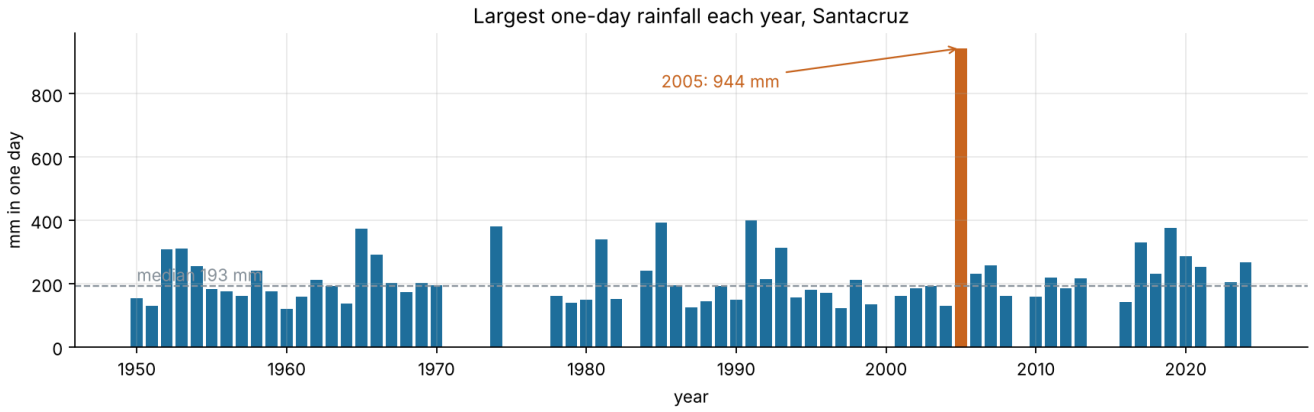
Rule	Why	Effect
Replace the 2005 storm with IMD's 944 mm	GHCN shows 461 mm on 26 July and 0.0 mm on 27 July. IMD reported 944 mm.	2005 annual maximum corrected from 461 to 944 mm
Coverage-adjust monsoon totals: mean daily rain × 122	A season with a few missing days would otherwise look drier than it was	Totals comparable across years
Keep values rounded to whole mm after 1973	A change in precision, not an error	No change

How we know 2005 was a recording error. GHCN dates each value by the end of IMD's 24-hour window (8:30 am to 8:30 am). We checked this on 30 August 2017: GHCN has 331.0 mm, and IMD reported 331.4 mm for 8:30 am 29 August to 8:30 am 30 August (India TV News). The 2005 storm began on the afternoon of 26 July, so its 24-hour total belongs on 27 July. GHCN's 0.0 mm for that day is a failed record. IMD's figure for the event is 944 mm (2005 Mumbai floods). We set 27 July 2005 to 944 mm and marked 26 July as missing. Without this check, the most important day in the dataset would have entered the analysis at half its true size.

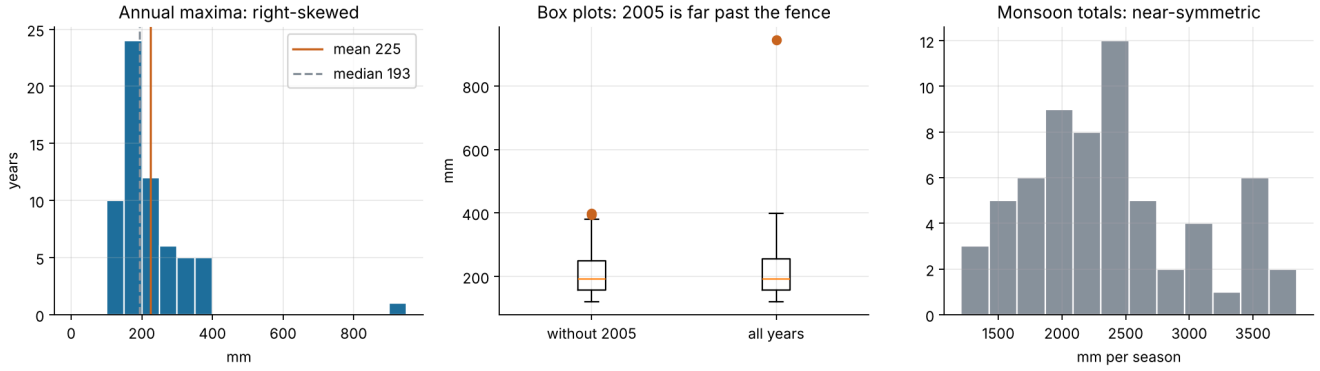
The heavy-rain thresholds follow IMD's categories: **very heavy rain ≥ 115.6 mm a day** and **extremely heavy rain ≥ 204.5 mm a day**.

4. Descriptive statistics and visualisation

Annual maxima are strongly right-skewed, and 2005 is an extreme outlier. Monsoon totals, by contrast, are close to symmetric. That contrast decides which distributions fit in Section 5.



Statistic	Annual max, all 63 years	Annual max, without 2005	Monsoon total (adjusted)
Mean	224.7 mm	213.1 mm	2,368 mm
Median	193.0 mm	192.8 mm	2,309 mm
Mode (50 mm class)	150–200 mm	150–200 mm	2,550–2,600 mm
Variance	13,954 mm²	5,564 mm²	415,555 mm²
Standard deviation	118.1 mm	74.6 mm	644.6 mm
Coefficient of variation	0.53	0.35	0.27
Q1 / Q3 (IQR)	157.9 / 254.5 (96.6)	157.9 / 250.1 (92.2)	1,938 / 2,592 (654)
Min / max	121.2 / 944.0	121.2 / 399.0	1,208 / 3,845
Skewness	3.90	1.04	0.53
Excess kurtosis	21.7	0.21	−0.39



- **Central tendency.** The mean annual maximum (225 mm) sits above the median (193 mm), the signature of a right-skewed variable. Removing one year moves the mean by 12 mm but the median by only 0.2 mm, so the median is the robust summary.
- **Spread.** One storm multiplies the variance by 2.5 (5,564 to 13,954 mm²). Skewness falls from 3.90 to 1.04 without 2005, but stays clearly positive, so the long right tail is not just one event.
- **Outliers.** The box-plot fence (Q3 + 1.5 × IQR) is 399.4 mm. Only 2005 lies beyond it, at 2.4 times the fence.
- **Counts.** Very heavy days (≥ 115.6 mm) per monsoon have mean 3.62, median 3, mode 3 and variance 4.56. A variance close to the mean is the first hint of a Poisson process.

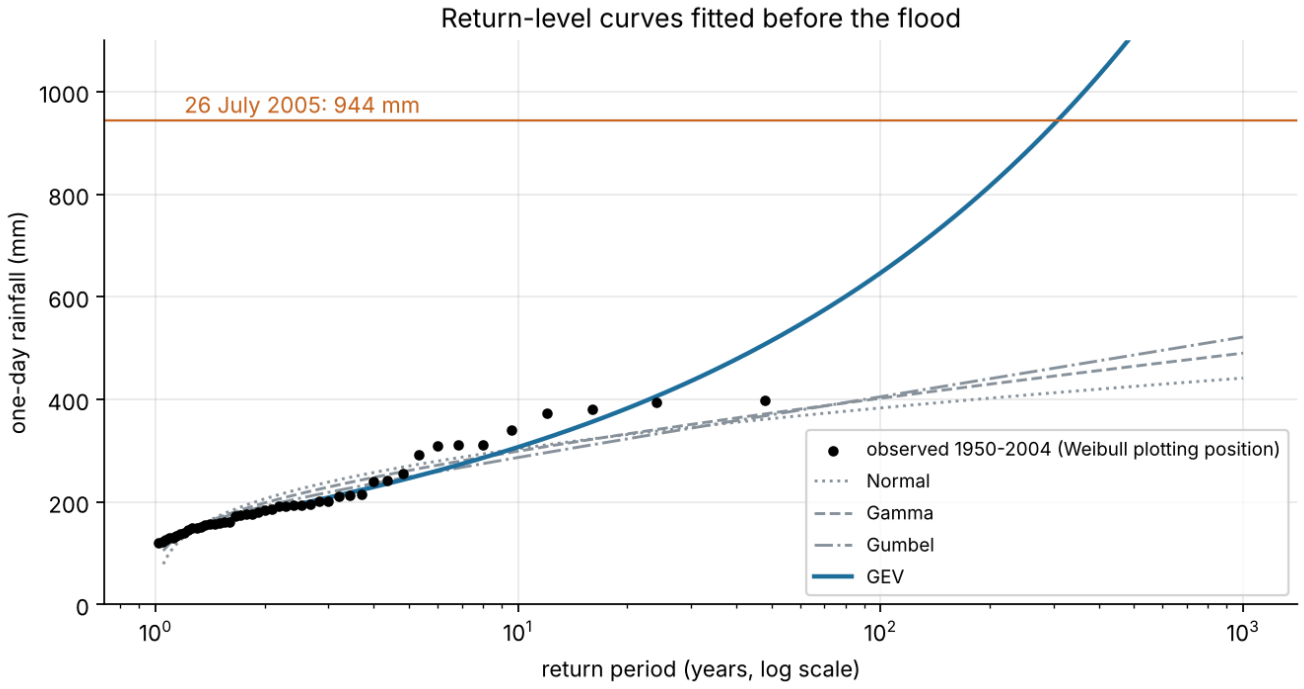
5. Random variables and probability distributions

The GEV distribution fits Mumbai's annual maxima best, and it is the only model that treats 944 mm as a possible event. Very heavy rain days follow a Poisson distribution, and detrended monsoon totals follow a Normal one.

5.1 Fitting the annual maximum, using only data before the flood

We fitted four models by maximum likelihood to the 47 complete years from 1950 to 2004, then asked each one how rare 944 mm would be. Goodness of fit uses a chi-square test with 6 equal-probability bins (expected count 7.8 per bin; df = 6 − 1 − number of fitted parameters). The Kolmogorov–Smirnov (KS) test and AIC are cross-checks; a lower AIC is better.

Model	χ^2	df	Critical (5%)	p (χ^2)	p (KS)	AIC	Verdict	100-yr level	Return period of 944 mm
Normal	15.43	3	7.81	0.001	0.032	544.7	Rejected	384 mm	5.4×10^{21} years
Gamma	7.77	3	7.81	0.051	0.146	532.8	Fits, barely	402 mm	1.7×10^{9} years
Gumbel	4.45	3	7.81	0.217	0.495	527.6	Fits	405 mm	4.3×10^{6} years
GEV	**0.87**	**2**	**5.99**	**0.647**	**0.946**	**522.8**	**Best fit**	**647 mm**	**308 years**



The three passing models agree in the middle of the data and disagree completely in the tail. That is why passing a chi-square test is not enough on its own. The GEV's shape parameter is **ξ = 0.36**, meaning a heavy, unbounded upper tail. In all 1,000 bootstrap

resamples ξ came out positive (95% CI 0.14 to 0.64), so the heavy tail is a property of the data, not of one sample.

GEV fitted on 1950–2004	Estimate	95% bootstrap CI
Location / scale	164.8 mm / 41.1 mm	—
Shape ξ	0.36	0.14 to 0.64
100-year one-day rainfall	647 mm	416 to 1,217 mm
Return period of 944 mm	308 years	65 to 12,299 years

What this means. A model fitted on data available in 2004 rated a 944 mm day as roughly a 1-in-300-year event, with a plausible range of 1-in-65 to 1-in-12,000. Rare, but well inside what infrastructure planning considers. The Gamma model used in the presentation rated it a 1-in-1.7-billion-year event, which would wrongly call it impossible.

5.2 The full record, 1950–2024

Refitting on all 63 years (8 bins, expected 7.9 per bin) gives the same ranking. Normal ($p < 0.001$) and Gamma ($p = 0.019$) are now rejected, Gumbel just passes ($p = 0.062$), and GEV fits best ($\chi^2 = 2.14$, $p = 0.71$; AIC 717.5). The website's calculator uses this fit: $\xi = 0.37$, location 172.9 mm, scale 47.3 mm.

Return period	GEV level	Gamma level
2 years	191 mm	213 mm
10 years	340 mm	345 mm
25 years	465 mm	404 mm
50 years	590 mm	446 mm
100 years	752 mm (95% CI 453 to 1,312)	485 mm
200 years	962 mm	523 mm

GEV and Gamma agree up to the 10-year level, then diverge sharply.

5.3 Binomial and geometric: what "100-year" really means

Treat each year as a Bernoulli trial with $p = 1/T$ of exceeding the T-year level. The waiting time to the first exceedance is geometric with mean T. The number of exceedances in n years is Binomial(n, p), so $P(\text{at least one}) = 1 - (1 - p)^n$.

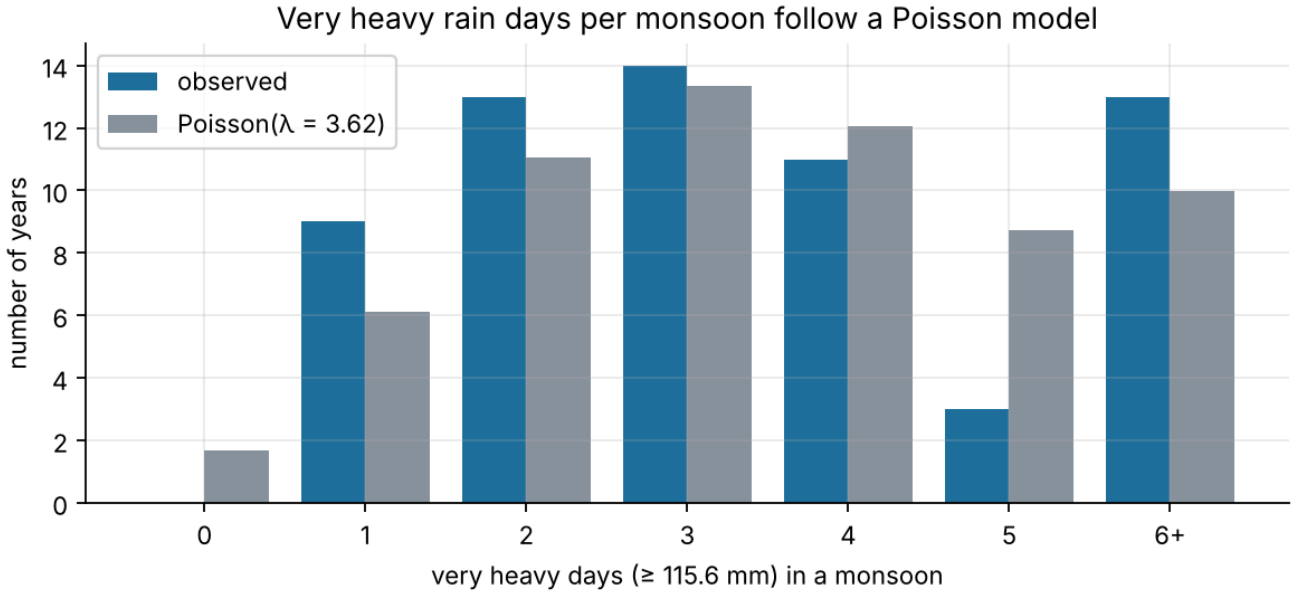
Years (n)	10-year event	50-year event	100-year event	300-year event
10	65.1%	18.3%	9.6%	3.3%
30	95.8%	45.5%	26.0%	9.5%
50	99.5%	63.6%	39.5%	15.4%
100	100.0%	86.7%	63.4%	28.4%

A 30-year home loan has a 26% chance of seeing a 100-year storm. Even a 2005-sized storm (about 1-in-300) has a 9.5% chance in that time.

We also checked the binomial mean and spread ($E[X] = np$, $SD = \sqrt{np(1-p)}$) against the 47 pre-2005 years. Observed exceedances of the GEV's 2-year, 10-year and 50-year levels were 24, 8 and 0, against expected 23.5, 4.7 and 0.94. All are within 1.6 SD, with exact binomial p of 1.00, 0.14 and 1.00. This is an in-sample check; the out-of-sample version is in Section 8.

5.4 Poisson: very heavy rain days per monsoon

The count of very heavy days (≥ 115.6 mm) in a fixed 122-day season is a count of rare events, so we model it as Poisson with λ = sample mean = **3.62 days per monsoon**.



- **Goodness of fit.** We merged the 0 and 1 classes so every expected count is at least 5. That gives $\chi^2 = 5.33$ on 4 df (critical 9.49), $p = 0.26$, so Poisson is not rejected.
- **Dispersion.** Variance divided by mean is 1.26 (test $p = 0.081$). That is mild over-dispersion: wet and dry monsoons cluster a little more than pure Poisson allows.

- **Predictions.** P(no very heavy day in a monsoon) = 2.7%; across 63 years there were none. P(5 or more) = 29.7%; observed 25%.
- **Limit.** For extremely heavy days (≥ 204.5 mm, λ = 0.75) the dispersion index is 1.56. Poisson then predicts at least one such day in 52.6% of monsoons against 41.3% observed. These rarer days cluster in a few very wet seasons, so a negative binomial model would suit them better.

5.5 Normal: where it does fit

The Normal model fails badly for annual maxima (Shapiro–Wilk W = 0.65, p < 0.001). A monsoon total, however, is a sum of 122 daily values, so the Central Limit Theorem applies. Raw totals narrowly fail (W = 0.957, p = 0.026) because they mix a drier era with a wetter one. After removing the trend (Section 7), the residuals are Normal (W = 0.984, p = 0.57).

- Mean monsoon total: 2,368 mm, 95% t-interval 2,206 to 2,530 mm (t = 2.00, df = 62).
- P(total > 3,000 mm) from the trend-adjusted Normal model: **7.0%** for a 1960 season and **34.0%** for a 2024 season. The observed shares are 6.5% in 1950–1987 and 34.4% in 1988–2024.

6. Hypothesis testing

Monsoons since 2010 are significantly wetter, but there is no significant change in the worst single day or in how often very heavy days occur. All tests use α = 0.05 and are two-sided unless stated. Periods are split at the midpoint: 1950–1987 (31 years) and 1988–2024 (32 years).

#	Question	H_0	Test	Statistic	p-value	Decision
H1a	Are the variances of annual maxima equal?	$\sigma_1^2 = \sigma_2^2$	F-test	F = 3.39 on (31, 30) df; critical 2.07	0.0012	Reject. Without 2005: F = 0.78, p = 0.50
H1b	Has the mean annual maximum changed?	$\mu_1 = \mu_2$ (213.9 vs 235.1 mm)	Welch t-test (variances unequal)	t = −0.72, df = 48.1	0.478	Fail to reject
H1c	Same, without assuming normality	Equal distributions	Mann–Whitney U	—	0.630	Fail to reject
H2	Do monsoon totals differ across four eras?	All era means equal	One-way ANOVA	F = 10.43 on (3, 59) df; critical 2.76	< 0.0001	**Reject**

#	Question	H_0	Test	Statistic	p-value	Decision
H2 check	Equal variances across eras?	Equal variances	Levene	—	0.939	ANOVA assumption holds
H2 check	Same, rank-based	Equal distributions	Kruskal–Wallis	—	0.0003	**Reject**
H3	Are very heavy days more frequent now? (one-sided)	$\lambda_2 \leq \lambda_1$ (3.55 vs 3.69 a season)	Two-sample Poisson z-test	z = 0.29; critical 1.645	0.386	Fail to reject
H4	Is the heavy-day class independent of period?	Independent	Chi-square test of independence, 2 × 3 table	χ^2 = 0.46, df = 2; critical 5.99	0.796	Fail to reject

Which eras differ (H2, Tukey HSD).

Era	Years	Mean monsoon total	SD
1950–69	20	2,327 mm	570 mm
1970–89	13	1,913 mm	432 mm
1990–2009	18	2,269 mm	543 mm
2010–24	12	3,077 mm	555 mm

The 2010–24 era is significantly wetter than each of the other three: +750 mm vs 1950–69 ($p = 0.002$), +1,164 mm vs 1970–89 ($p < 0.001$) and +808 mm vs 1990–2009 ($p < 0.001$). No other pair differs.

Reading the results.

- One storm changes the F-test verdict. With 2005 the variances differ ($p = 0.001$); without it they do not ($p = 0.50$). A single outlier can decide a test, so we report both versions.
- "Not significant" does not mean "no change". Section 7 shows that this record could only detect a trend in annual maxima larger than 19 mm per decade with 80% power. Smaller real trends, like the ones regional studies report, would most likely be missed. That would be a **Type II error**, and it is likely here.

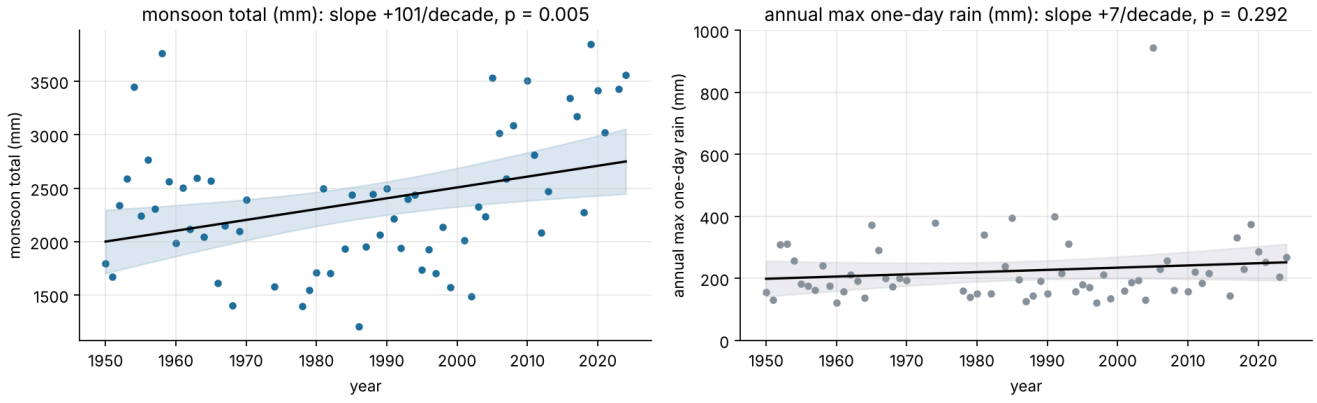
7. Regression analysis

Monsoon totals rise by about 100 mm per decade. A wetter monsoon packed into fewer rainy days produces a worse single day. The annual maximum itself shows no significant linear trend.

7.1 Simple linear trends

Each series is regressed on year: $y = \beta_0 + \beta_1 \cdot \text{year}$. H_0: $\beta_1 = 0$ is tested with a t-test on $n - 2$ df. The Mann–Kendall rank test is a cross-check that does not assume linearity or normality. The last column is the smallest trend the record could detect with 80% power.

Series	Slope per decade (95% CI)	t	p	R^2	Mann–Kendall p	Detectable at 80% power
Annual max	+7.2 mm (−6.3 to +20.6)	1.06	0.292	0.018	0.231	19.1 mm
Annual max without 2005	+2.7 mm (−5.9 to +11.4)	0.64	0.526	0.007	0.302	12.2 mm
Monsoon total	**+101 mm (+32 to +171)**	**2.91**	**0.005**	0.122	**0.017**	98.8 mm
Very heavy days	+0.10 days (−0.15 to +0.34)	0.82	0.418	0.011	0.280	0.35 days
Wet days (≥ 2.5 mm)	+0.32 days (−0.92 to +1.55)	0.51	0.610	0.004	0.377	1.75 days



Checking the monsoon-total model. The fitted line is total = −17,730 + 10.12 × year.

- The residuals are Normal (Shapiro p = 0.57).
- The Durbin–Watson statistic is 1.17, below the ideal 2. Neighbouring monsoons are positively correlated, which makes ordinary p-values too optimistic.
- With Newey–West (HAC) standard errors that allow for this correlation, the slope test gives t = 1.92 and p = 0.054.

The honest summary: the rise is strongly supported as a difference between eras (ANOVA, Section 6), and borderline as a steady linear trend.

Prediction. The fitted mean monsoon total for 2030 is 2,810 mm (95% CI for the mean 2,470 to 3,150 mm). The 95% prediction interval for a single 2030 season is much wider, 1,546 to 4,074 mm, because one season varies far more than the average does.

7.2 Multiple regression: what drives the worst day?

We left 2005 out because one 944 mm point would dominate the fit; its Cook's distance is 0.88. The model, on 62 years:

$\text{max} = \beta_0 + \beta_1 \cdot \text{year} + \beta_2 \cdot (\text{monsoon total} / 100\ \text{mm}) + \beta_3 \cdot \text{wet days} + \varepsilon$

Term	Coefficient	SE	t	p
Intercept	895.5	855.2	1.05	0.299
Year	−0.33 mm per year	0.43	−0.77	0.445
Monsoon total (per 100 mm)	**+7.25 mm**	2.00	3.62	**0.001**
Wet days	**−2.78 mm per extra day**	1.11	−2.52	**0.015**

$R^2 = 0.19$ (adjusted 0.15). Overall $F = 4.54$, $p = 0.006$. VIFs are 1.18 to 2.06, so the predictors are not collinear.

Holding the season total fixed, every extra wet day lowers the expected annual maximum by about 2.8 mm. The same rain spread over fewer days means heavier individual days. Once the season total is accounted for, year has no effect. So any rise in extremes runs through wetter monsoons, not a separate time trend. R^2 is low (0.19): most of the variation in a year's worst day is the randomness the GEV model describes. The residuals are slightly skewed (Shapiro $p = 0.018$), so treat the p-values as approximate.

8. Validation, errors and deviations

On years it never saw, the GEV model stays within its stated uncertainty and the Poisson model holds. The linear trend does not extrapolate. We fitted everything on 1950–1990 (34 complete years) and tested it on 1991–2024 (29 complete years).

Check	Trained on 1950–1990	Predicted for 1991–2024	Observed	P(this many or more)	Verdict
GEV 10-year level	314 mm	2.9 years above	4	0.33	Consistent

Check	Trained on 1950–1990	Predicted for 1991–2024	Observed	P(this many or more)	Verdict
GEV 50-year level	530 mm	0.58 years above	1 (2005)	0.44	Consistent
Gamma 10-year level	304 mm	2.9 years above	5	0.16	Too many, not yet significant
Gamma 50-year level	379 mm	0.58 years above	2	0.11	Too many, not yet significant
Poisson, very heavy days	$\lambda = 3.45$ per season	100 days	111	0.14	Consistent (+10%)
Linear trend, monsoon total	Line through 1950–1990	RMSE 1,248 mm	No-trend mean: RMSE 793 mm	—	Trend fails out of sample

- **Return levels.** Gamma's tail is too light. It was exceeded more often than predicted at both levels, while GEV stayed close to expectation. With only 29 test years neither gap is statistically significant, but the direction matches Section 5: Gamma underestimates extremes.
- **Poisson.** The model predicted 100 very heavy days and the test period produced 111. That is within Poisson noise ($p = 0.14$).
- **Regression.** The 1950–1990 data contain the dry 1970s and 1980s, so their fitted line pointed downward. The 1991–2024 seasons, especially after 2010, went the other way. A straight line fitted to one era is a poor forecast for the next. Regression describes the data it was fitted on; extrapolating it decades ahead is risky.

Sources of error.

1. **Missing days.** Thirteen seasons failed the 85% rule. Seasons we kept can still miss a storm on an unrecorded day, so heavy-day counts lean slightly low.
2. **Recording failures.** The 2005 storm was stored as 461 mm on the wrong day. Without cross-checking against IMD, the key observation would have been wrong by 483 mm.
3. **Precision change.** Many values after 1973 are rounded to whole millimetres. This is negligible for extremes of hundreds of millimetres.
4. **One gauge.** A single station can miss a storm centred a few kilometres away. On 26 July 2005, Colaba, also in Mumbai, recorded only about 74 mm.
5. **Model assumptions.** The binomial and return-period arithmetic assume independent years with a constant probability. The wetter post-2010 era and the autocorrelation in Section 7 show both assumptions hold only approximately.
6. **Short record.** 47 to 63 years cannot pin down a 100-year level. That is why every tail estimate carries a bootstrap interval, and why those intervals are wide.

9. How this report extends the presentation

The presentation's question and conclusion stand: probability gives the odds, not the date. This report reruns every calculation on the full cleaned record, so some figures change. Where a number below differs from a slide, the notebook value is the reproducible one.

Slide	Presentation	This report	Why it differs
Step 1: which distribution?	33 years, 1971–2004. Gamma fits ($\chi^2 = 1.39$); Normal and Weibull rejected	47 complete years, 1950–2004. Normal rejected; Gamma passes narrowly ($p = 0.051$); **GEV fits best** ($\chi^2 = 0.87$, $p = 0.65$, lowest AIC)	Longer record; extreme-value models (Gumbel, GEV) added to the comparison
Step 1 chart note	Record before 2005 was 373 mm (1965)	Record before 2005 was 399 mm (1991); 373 mm was the record up to 1973	Full record includes 1974, 1985 and 1991 storms above 373 mm
Step 2: binomial table	9.6%, 26.0%, 39.5%, 63.4%	Same	Arithmetic confirmed
Step 3: Gamma fit	$k = 12.3$, $\theta = 16.3$; 944 mm ≈ 1 in 10 trillion years	$k = 8.71$, $\theta = 23.7$; 944 mm ≈ 1 in 1.7 billion years. **GEV: 1 in 308 years**	Different sample; the GEV result shows the "impossible" verdict comes from Gamma's light tail
Step 3: 95% CI for the mean	179 to 223 mm	183.6 to 228.8 mm ($n = 47$, $t = 2.013$)	Different sample
Step 4: expected vs observed	16.5/13, 3.3/5, 0.66/1	23.5/24, 4.7/8, 0.94/0 (GEV levels, 47 years)	Different sample and model
Step 5: regression	$\hat{y} = 1302.8 - 0.56 \times$ year, $p = 0.37$; $F = 1.63$ vs critical 3.23	Annual max: +7.2 mm per decade, $p = 0.29$. Welch t $p = 0.48$. Variance-ratio F critical ≈ 2.07	The slide's line implies a mean year of about 1968, which does not fit a 1971–2004 sample. 3.23 is the critical value for $F(2, 40)$, not for a two-group variance ratio

Slide	Presentation	This report	Why it differs
Step 5: Roxy et al.	Widespread extremes tripled "since 1971"	Threefold rise over **1950–2015**, central India (Roxy et al. 2017)	Period as stated in the paper
Step 1 data note	Santacruz 2005 value from IMD	GHCN stores 461 mm; corrected to IMD's 944 mm	Documented recording error (Section 3)

The biggest change is in the conclusion about "limits". The model did not miss 944 mm because probability fails. It missed it because Gamma's tail is too light for rainfall extremes. With the right family, data available in 2004 rated a 944 mm day as rare but plannable.

10. The product: flood-odds website

The website turns the fitted models into a calculator anyone can use, so the analysis becomes a working tool rather than a one-off result. It is a static site in the project repository's `docs/` folder and runs on GitHub Pages with no server or build step.

Feature	What the user does	Statistics behind it
Flood-odds calculator	Picks a one-day rainfall (120–1,200 mm, with 2005, 2017 and 2019 presets) and a time horizon (1–100 years)	Annual exceedance probability $p = 1 - F(x)$, return period $1/p$, Binomial P(at least one in n years) $= 1 - (1 - p)^n$, expected count np
Model switch	Compares GEV, Gumbel, Gamma and Normal, fitted on 1950–2024 or on 1950–2004 only	Maximum-likelihood parameters exported from the notebook
Live annual-maxima chart	Sees which years reached the chosen threshold	Observed frequency, to compare with the model's p
Return-level chart	Compares the four models' tails against 944 mm	Weibull plotting positions and model quantile curves
Poisson panel	Picks k and reads P(at least k very heavy days in a monsoon)	Poisson($\lambda = 3.62$) against the observed share

The browser code reimplements each distribution function: the GEV and Gumbel CDFs, the regularised incomplete gamma function and the Normal tail. Its outputs were checked against SciPy for all eight fitted models and match to at least six significant figures.

Example: 944 mm returns 307.6 years under the 1950–2004 GEV in both.

The site states its limits on the page. It estimates the chance of rainfall at one gauge, not whether a street floods.

11. Conclusion and limitations

Probability could not have predicted the date of the 2005 flood, but it could have put a number on the risk. That number depended entirely on choosing the right distribution.

Hypothesis	Result
H-dist: annual maxima are heavy-tailed, not Normal	**Supported.** GEV fits best ($\xi = 0.36 > 0$ in every bootstrap resample). Normal is rejected.
H-count: very heavy days are Poisson	**Supported**, with mild over-dispersion (χ^2 $p = 0.26$; variance/mean 1.26)
H-change: rainfall has changed	**Partly.** Monsoon totals rose, with 2010–24 wetter than every earlier era (ANOVA $p < 0.001$). No significant change in the annual maximum or heavy-day frequency, but the record has low power to detect one.
H-predict: early-trained models predict later years	**Supported for GEV and Poisson. Not supported for the linear trend**, which failed out of sample.

Limitations.

- **One gauge.** Results describe Santacruz only.
- **Short record.** 47 to 63 years leave wide intervals on every tail estimate.
- **Stationarity.** The models assume independent years with constant risk, while monsoon totals are shifting.
- **Data quality.** The 2005 error shows the raw record must be cross-checked, and other unverified errors may remain.
- **Rainfall is not flooding.** Hourly intensity, tides and drainage capacity decide whether 300 mm becomes a flood.

Next steps.

- Add Colaba and other IMD stations, which raises statistical power.
- Use a peaks-over-threshold (Generalised Pareto) model on daily data.

- Let GEV parameters vary with year to model the changing climate directly.

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Project code, data and figures: `analysis/flood_analysis.ipynb` in the project repository.